

Saarthi

A voice-native product for guided financial applications



Hackathon Track: PS2 (Build a voice-native product)



Team Name: **First Principles**



Team Members

Pramit De — Team Leader

Deepmalya Rakshit

Anita's problem

27.18%

of respondents met the minimum thresholds across financial knowledge, behaviour, and attitude.

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Borrowers need a way to clarify a financial term and then continue answering the same question.

Comparison journey study

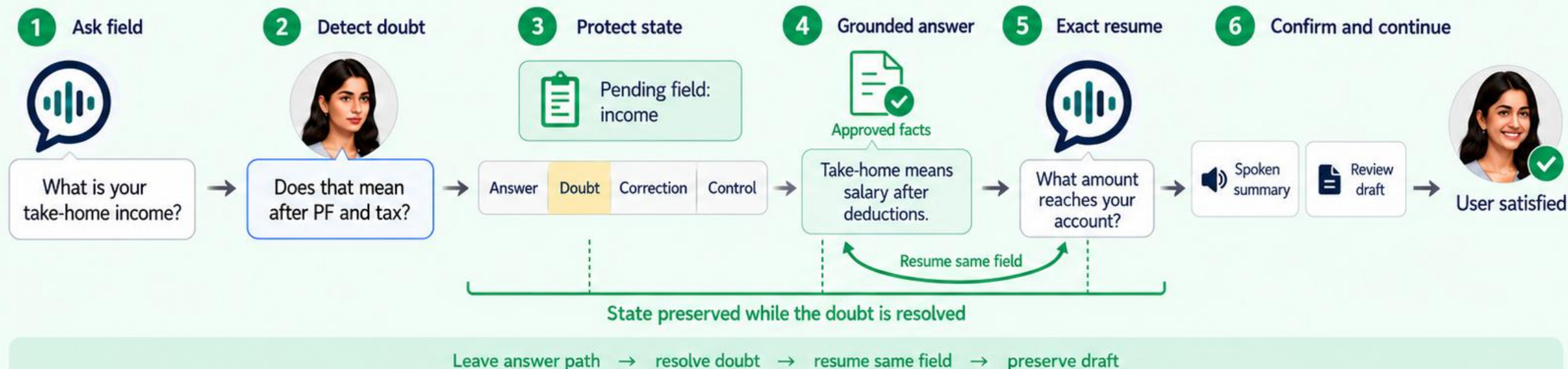
Rigid IVR failure vs doubt-aware voice agent recovery

✗ (a) Rigid form / IVR: failure

⚠ Contextual doubt mishandled

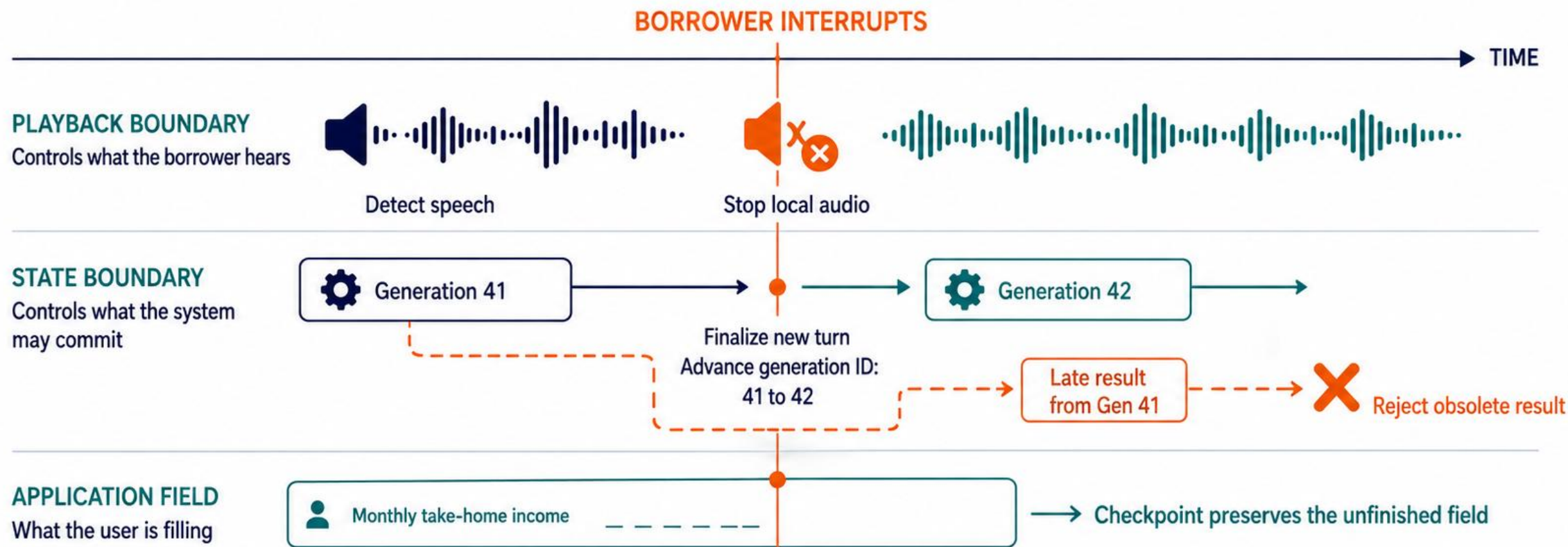


✓ (b) Saarthi journey

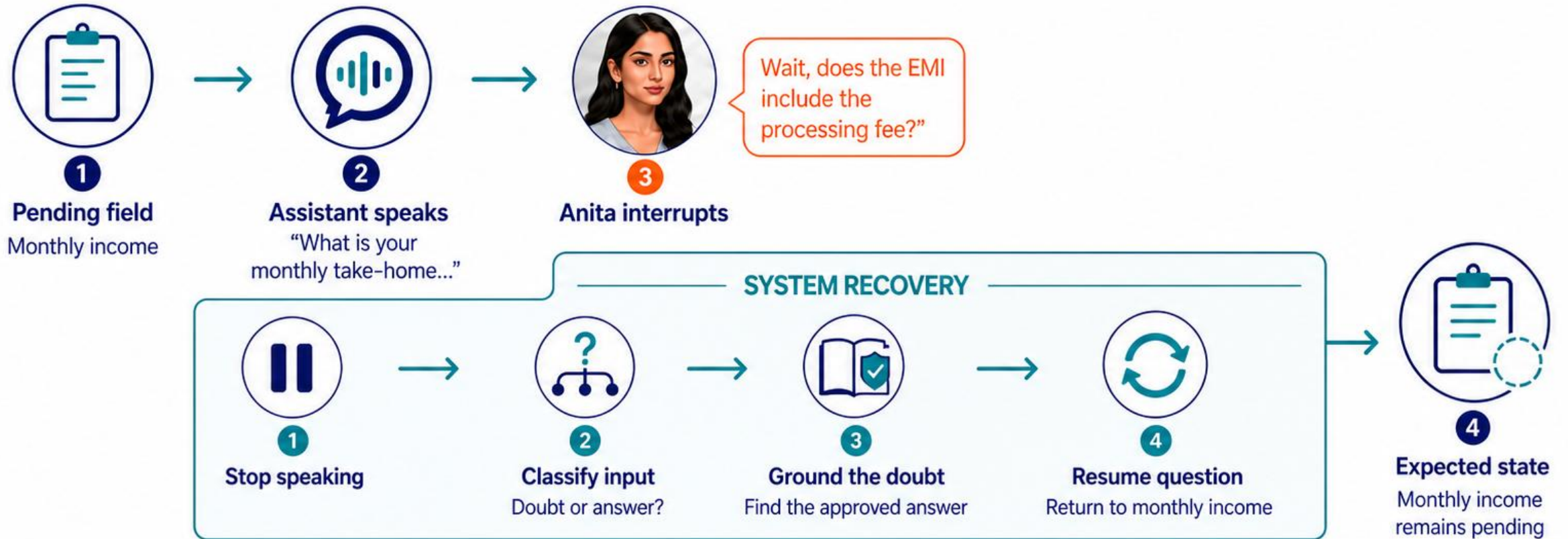


Interruption and state-consistent recovery

How do we stop speech immediately without letting stale work corrupt the application state?



The deliberate failure case



Safety invariants



The doubt must not become an income answer.



Confirmed fields must remain unchanged.



The same income question must resume.

Measured Result: Every claim leaves a trace

APPEND-ONLY EVENT TRACE

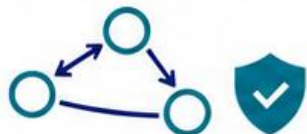


Material decisions remain inspectable across the conversation.

STATE INTEGRITY

Zero unintended mutations

30 scripted conversations
71 transitions per approach



FINANCIAL FACTS

45 of 45 facts preserved

Listener-oriented renderer

Synthesized audio  97.8%

Ordinary prose  15.6%

Bounded synthesized-audio evaluation

CANCEL-FIRST RECOVERY

No stale re-entry

12 cancellation scenarios

17.5 ms p95

Application-side cancellation path

Four controlled TR-5 stop trials



RESPONSE RELEASE p95

Normal/control response  104.4 ms

Grounded/product-question  3,301.5 ms

17.5 ms is a controlled application-side measurement, not real-world acoustic end-to-end latency.

A borrower can interrupt, resolve a grounded doubt, and resume the correct field without losing control of the draft.

Synthetic engineering results · not a production user study